CPPC 21/L

DIGITAL SIGNAL PROCESSING

ACTIVITY No.4

Face and Person Detection

Submitted by:

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Submitted to:

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I. Objectives

- To detect human faces using MATLAB and a webcam.
- To detect a person or human body figure through MATLAB.
- To use real-time image capture for computer vision processing.
- To apply built-in MATLAB functions for object detection.
- To understand how bounding boxes are used to highlight detected objects.

II. Materials

- MATLAB
- Laptop
- Webcam
- Computer Vision Toolbox

III. Procedures

1. Open MATLAB

- Launch the MATLAB application.
- Click New Script to create a new script file.
- Make sure that the webcam is properly connected and working.
- Ensure that the Computer Vision Toolbox is available in MATLAB.

2. Run Face Detection

- Create a new MATLAB script for face detection.
- Copy or type the face detection code in the script.
- The code uses the webcam to capture real-time images.
- It also uses a cascade object detector to locate faces.
- Once a face is detected, a rectangle or bounding box will appear around it.
- Save the file as:
    - LOGATOC_faceDetection.m

- Click Run to start the face detection process.

```
clc;
clear all;

cam = webcam;
img = snapshot(cam);

faceDetector = vision.CascadeObjectDetector();

imshow(img);

while true
    img = snapshot(cam);
    grayImg = rgb2gray(img);

    bbox = step(faceDetector, grayImg);

    detectedFace = insertObjectAnnotation(img, 'rectangle',
bbox, 'Face');

    imshow(detectedFace);
end
```

Note on Image-Based Detection

- During the initial testing of the face detection code, an error occurred at the line cam = webcam;.
- The error happened because the local MATLAB installation could not access the webcam function properly.
- The MATLAB Add-On Explorer was opened to install the needed webcam support package, but it required an active MathWorks Software Maintenance license.
- Because of this limitation, image-based face detection was temporarily used as an alternative method.
- The same face detection concept was still applied because MATLAB detected a face from an image and highlighted it using a bounding box.
- The real-time webcam-based face detection and person detection will still be shown using Online MATLAB or another MATLAB setup that supports webcam access.

3. Run Person Detection

- Create another MATLAB script for person detection.
- Copy or type the person detection code in the script.

- The code uses the webcam to capture real-time images.
- It applies the peopleDetectorACF function to detect a human figure.
- If a person is detected, a bounding box with a label will be displayed.
- Save the file as:
    - PersonDetection_Miras.m
- Click Run to start the person detection process.

```
clc;
clear;

cam = webcam;

personDetector = peopleDetectorACF();

while true
    img = snapshot(cam);

    [boxes, scores] = detect(personDetector, img);

    selectedBox = scores > 20;
    validBoxes = boxes(selectedBox, :);

    if ~isempty(validBoxes)
        detectedPerson = insertObjectAnnotation(img, 'rectangle',
validBoxes, 'Person');
    else
        detectedPerson = img;
    end

    imshow(detectedPerson);
    drawnow;
end
```

IV. Conclusion

The activity showed how MATLAB can be used to perform real-time face and person detection using a webcam. By running the face detection and person detection scripts, the system was able to capture live images and identify the target objects through bounding boxes. This activity demonstrated the basic use of computer vision in detecting human faces and body figures, showing how MATLAB can be applied in real-time image analysis and object detection.

V. Documentation

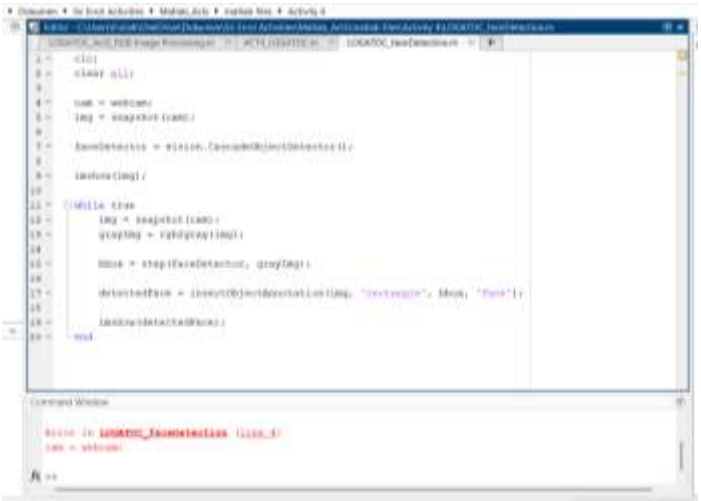


Figure 1. Error in Webcam-Based Face Detection

This screenshot shows the error encountered when running the original face detection code. The error occurred at the command cam = webcam;, which means that MATLAB could not access the webcam function.

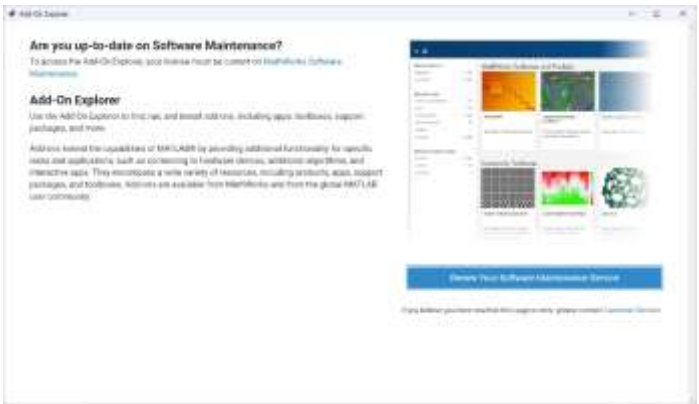
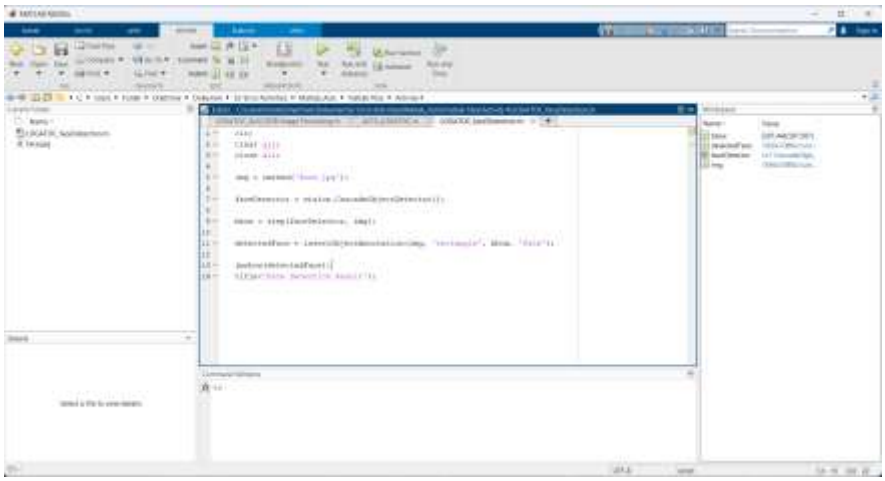


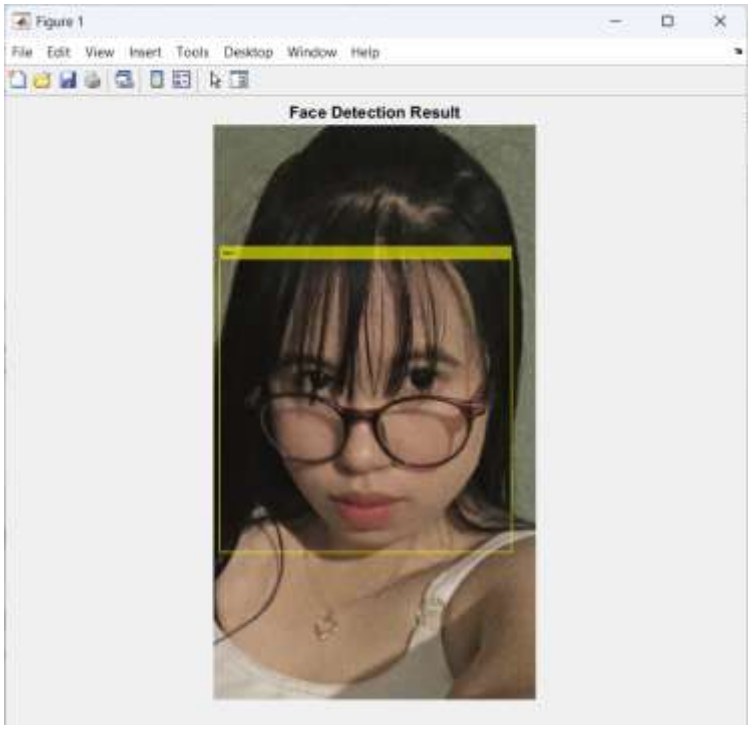
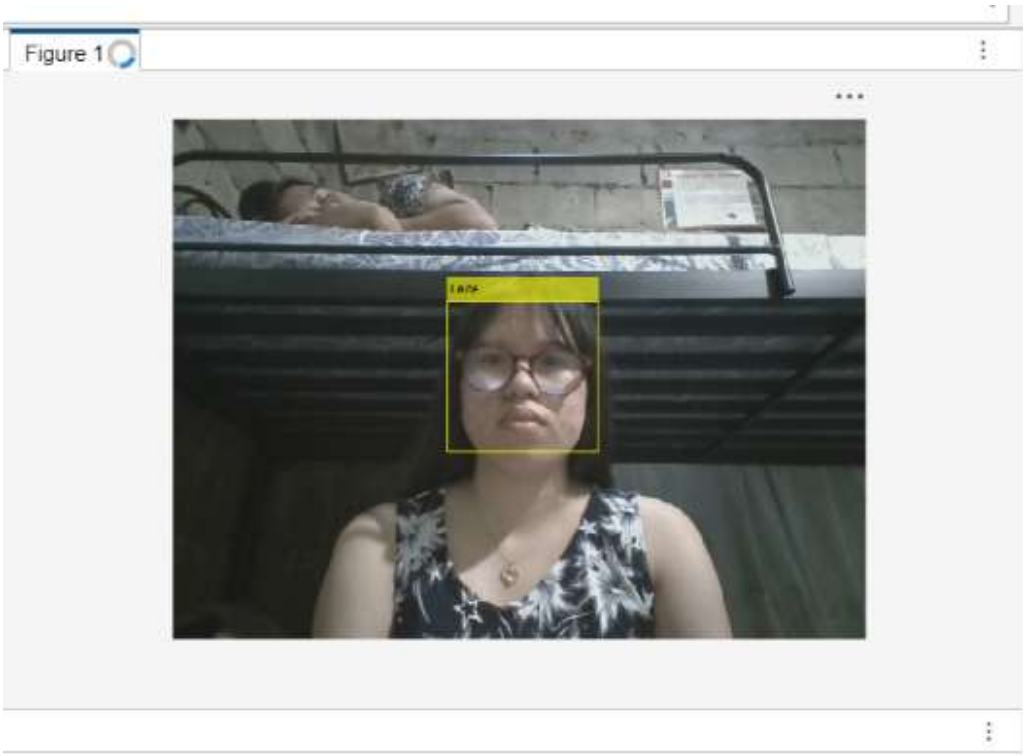
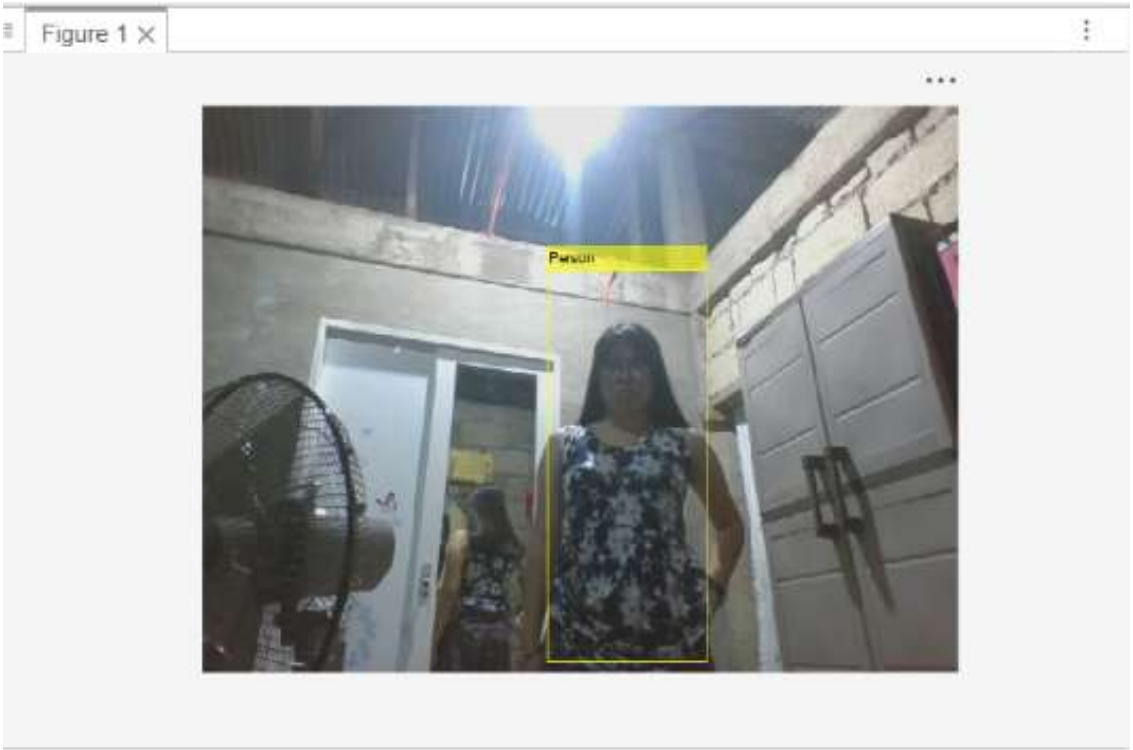
Figure 1. Error in Webcam-Based Face Detection

This screenshot shows that MATLAB Add-On Explorer could not be accessed because the software requires an active MathWorks Software Maintenance license. This prevented the installation of the required webcam support package.

Image-Based Face Detection



Result:



MATLAB Online